

Madhya Pradesh – Grade 10

2016 – Mathematics

Choose the correct option and write it in your answer book:

1. The system of equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ represent two parallel lines if,

(a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(b) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

(c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(d) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Answer: (c)

2. The third proportional to 9, 12 is:

(a) $6\sqrt{3}$

(b) $3\sqrt{6}$

(c) $\frac{27}{4}$

(d) 16

Answer: (d)

Solution: $9 : 12 :: 12 : x$

$$\frac{9}{12} = \frac{12}{x}$$

$$x = \frac{144}{9} = 16$$

3. The maximum degree of variable used in quadratic equation is:

(a) 1

(b) 2

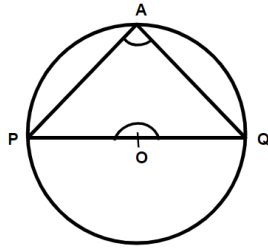
(c) 3

(d) 4

Answer: (b)

Solution: highest degree in quadratic is always 2.

4. The measure of $\angle PAQ$ in the figure given below is:



- (a) 45°
- (b) 180°
- (c) 90°
- (d) 60°

Answer: (c)

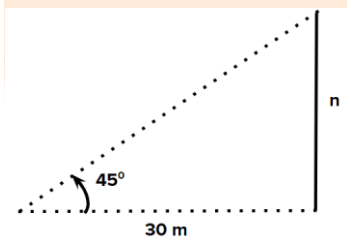
Solution: semicircle angle is 90° .

5. From a point 30m away from the foot the building the angle of elevation of the top of building is 45° . The height of the building will be:

- (a) 25 m
- (b) 30 m
- (c) $25\sqrt{2}$ m
- (d) $30\sqrt{2}$ m

Answer: (b)

Solution:



$$\tan 45 = \frac{h}{30}$$

$$h = 30m$$

Fill in the blanks:

1. The zero of the linear polynomial $ax + b$ is

Answer: $-\frac{b}{a}$

Solution:

$$ax + b = 0$$

$$x = -\frac{b}{a}$$

2. The rate of depreciation is

Answer: negative

3. If the sides of two triangles are proportional, then triangles will be

Answer: similar

Solution: In similarity the corresponding sides of two triangles are always proportionl

4. The radius of circle is 7 cm, then its area will be

Answer: 154 sq.cm

Solution:

$$\text{Area} = \pi r^2$$

$$= \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

5. The length of the diagonal of cube is $12\sqrt{3}$ cm. The length of the edge of the cube will be

Answer: 12 cm

Solution:

$$\text{Diagonal} = \sqrt{l^2 + l^2 + l^2}$$

$$12\sqrt{3} = \sqrt{3}l$$

$$l = 12 \text{ cm}$$

Match the correct column:

'A'	'B'
(i) $\sin^2 25^\circ + \cos^2 25^\circ$	(a) $\cos$
(ii) $1 + \cot^2 \theta$	(b) 1
(iii) $\sin(90^\circ - \theta)$	(c) $\text{cosec}^2 \theta$
(iv) $\sec 60^\circ$	(d) $\tan 41^\circ$
(v) $\tan 45^\circ$	(e) 2
	(f) $\frac{1}{2}$
	(g) $\cot 41^\circ$

Answers:

(i) – b

(ii) – c

(iii) – a

(iv) – e

(v) – g

Write True/False in the following:

1. $x + 2\sqrt{x}$ is not polynomial.

Answer: True

Solution: Polynomial degree should be whole number only not fraction.

2. Income tax is a direct tax.

Answer: True

3. Angles in the same segment of a circle are equal.

Answer: True

4. The volume of hemisphere is $3\pi r^2$

Answer: False

Solution: Volume = $\frac{2}{3}\pi r^3$

5. The probability of definite event is always one.

Answer: True

Write the answer in one word/sentence of each:

1. In equation $x + 2y = 5$, if $y = 0$ then write the value of x .

Answer: 5

Solution: $x + 2y = 5$

$$x + 2 \times 0 = 5$$

$$x = 5$$

2. Write the formula of Heron to find the area of a triangle.

Answer: $\sqrt{s(s-a)(s-b)(s-c)}$

3. Write the statement of Pythagoras theorem.

Answer: In a right-angle triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

4. Write the number of circles passing through three non-collinear points.

Answer: one

5. Write the value of the mode of the following data:

2, 3, 4, 2, 12, 9, 6, 9, 5, 9.

Answer: 9

Question: Write the statement of basic proportionality theorem (Thales theorem)

Or

Write the property of angle-angle similarity.

Solution:

(a) BPT theorem states that, if a line parallel to a side of a triangle intersect the other sides into two distinct points, then the line divides those sides in equal proportion.

(b) In two triangles, if two pairs of corresponding angles are equal, then it can be shown that all three pairs of corresponding angles are equal, by angle sum property.

Question: The perimeters of two similar triangle are 30 cm and 20 cm respectively. If one side of one triangle is 12 cm, find the corresponding side of other triangle.

Or

Triangle ACB is an isosceles triangle such that $AC=BC$. If $AB^2 = 2AC^2$. Prove that $\triangle ACB$ is a right-angled triangle.

Solution:

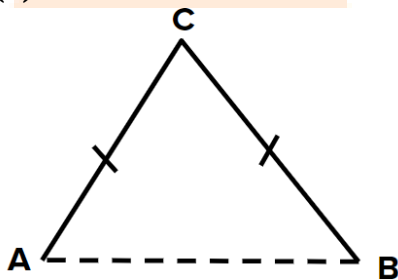
(a) If two triangle are similar then ratio of the perimeters of the triangles is equal to the ratio of their corresponding sides.

Let the required side be x following the theorem,

$$\frac{30}{20} = \frac{12}{x}$$

$$x = \frac{20 \times 12}{30} = 8 \text{ cm}$$

(b)



Given $AC = BC$

$$AB^2 = 2AC^2$$

To Prove: $\triangle ACB$ is right angled

Proof: let $\triangle ACB$ is right angled at C

$$\text{So, } AC^2 + BC^2 = AB^2$$

$$\therefore AC = BC \text{ and } AB^2 = 2AC^2$$

$$AC^2 + AC^2 = 2AC^2$$

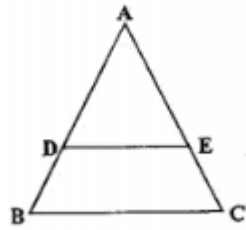
$$2AC^2 = 2AC^2$$

$$\text{L.H.S} = \text{R.H.S}$$

Question: Triangle ABC and triangle PQR are two similar triangles. The areas of these are 64 cm^2 and 100 cm^2 respectively. If $QR = 12 \text{ cm}$. Then find value of side BC.

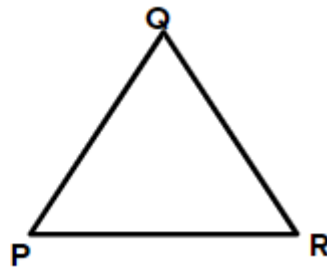
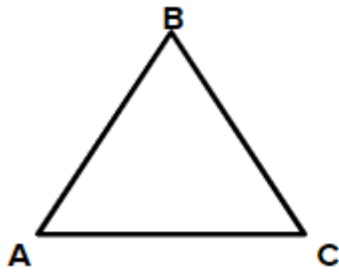
Or

In the figure given below $DE \parallel BC$, If $\frac{AD}{DB} = \frac{3}{5}$ and side $AC = 6 \text{ cm}$. Then find the value of AE.



Solution:

(a)



$$\triangle ABC \sim \triangle PQR$$

Area ratio:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

$$\frac{\text{Ar.}\triangle ABC}{\text{Ar.}\triangle PQR} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$$

$$\frac{64}{100} = \left(\frac{BC}{12} \right)^2$$

$$\frac{8}{10} = \frac{BC}{12}$$

$$BC = \frac{96}{10} = 9.6 \text{ cm}$$

(b)

$\therefore DE \parallel BC$

Using B. P. T theorem

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$EC = AC - AE$$

$$\frac{AD}{DB} = \frac{AE}{AC - AE}$$

$$\frac{3}{5} = \frac{AE}{6 - AE}$$

$$18 - 3AE = 5AE$$

$$18 = 8AE$$

$$AE = \frac{18}{8} = 2.25 \text{ cm}$$

Question: Write any two properties of Arithmetic mean.

Or

Find the median of the following value : 5, 10, 3, 7, 1, 9, 6, 2, 11.

Solution:

(a)

(i) The mean of n observation $x_1, x_2, \dots, x_n$ is $\bar{x}$. If each observation is decreased by p, the mean of the new observation is $(\bar{x} - p)$

(ii) The mean of n observation $x_1, x_2, \dots, x_n$ is $\bar{x}$. If each observation is increased by p, the mean of the new observation is $(\bar{x} + p)$

(b)

Arrange the data

1, 2, 3, 5, 6, 7, 9, 10, 11

Median = $\left(\frac{n+1}{2}\right)$ th observation for odd

$$= \frac{9+1}{2} = \frac{10}{2} = 5^{\text{th}} \text{ observation}$$

Median = 6

Question: Write the probability of getting an odd number in a single throw of die.

Or

If two coins are tossed simultaneously, find the probability of getting two heads.

Solution:

(a)

Sample space of die = {1, 2, 3, 4, 5, 6}

Odd event = {1, 3, 5}

Probability of odd = $\frac{\text{oddevent}}{\text{sample space}}$

$$= \frac{3}{6} = \frac{1}{2}$$

(b)

Sample space of two coin = {HH, HT, TH, TT}

Two heads event = {HH}

$$\begin{aligned}\text{Probability} &= \frac{\text{event}}{\text{sample space}} \\ &= \frac{1}{4}\end{aligned}$$

Question: Solve the following system of equation:

$$\begin{aligned}3x - 2y &= 4 \\ y + 2x &= 5\end{aligned}$$

Or

Find those values of m, for which the system

$$2x + my - 4 = 0$$

$$3x - 7y - 10 = 0 \text{ has}$$

(i) a unique solution, (ii) no solution.

Solution:

(a) $3x - 2y = 4 \dots\dots\dots(1)$

$2x + y = 5 \dots\dots\dots(2)$

Multiply equation 2 by 2 and add

$$3x - 2y = 4$$

$$4x + 2y = 10$$

$$7x = 14$$

$$x = 2$$

From equation (2) $y = 5 - 2x$

$$= 5 - 2 \times 2$$

$$= 5 - 4$$

$$y = 1$$

(b)

$$2x + my - 4 = 0$$

$$3x - 7y - 10 = 0$$

Comparing with $ax + by + c = 0$

$$a_1 = 2, b_1 = m, c_1 = -4$$

$$a_2 = 3, b_2 = -7, c_2 = -10$$

(1) for unique solution,

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\frac{2}{3} \neq \frac{m}{-7}$$

$$-\frac{14}{3} \neq m$$

For all real value of m , $\left(m \neq \frac{-14}{3}\right)$ the system has unique solution.

(2) for no solution the lines are parallel such that

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{2}{3} = \frac{m}{-7} \neq \frac{-4}{-10}$$

$$m \neq \frac{-7 \times 4}{10} = \frac{-14}{5}$$

So, m takes every value except $-\frac{14}{5}$

Question: The sum of two number is 7. If the sum of these number is seven times of its difference. Find the number.

Solution:

Let the number be x and y

$$x + y = 7 \dots\dots\dots (1)$$

$$x + y = 7(x-y) \dots\dots (2)$$

$$x + y = 7x - 7y$$

$$0 = 6x - 8y$$

$$0 = 3x - 4y \dots\dots\dots (3)$$

Solving (1) and (3)

$$x + y = 7 \times 4$$

$$3x - 4y = 0$$

$$4x + 4y = 28$$

$$7x = 28$$

$$x = 4$$

$$y = 7 - x \text{ (from equation 1)}$$

$$= 7 - 4 = 3$$

Question: If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b}$, then prove that $(b-c)x + (c-a)y + (a-b)z = 0$.

Or

What should be subtracted from 11, 20, 26, and 50 so that the remaining numbers are in proportion?

Solution:

(a)

$$\frac{x}{(b+c)} = \frac{y}{(c+a)} = \frac{z}{(a+b)} = k \text{ (let)}$$

$$x = k(b+c)$$

$$y = k(c+a)$$

$$z = k(a+b)$$

$$\text{L.H.S } (b-c)x + (c-a)y + (a-b)z$$

$$= (b-c)k(b+c) + (c-a)k(c+a) + (a-b)k(a+b)$$

$$= k(b^2 - c^2) + (c^2 - a^2)k + k(a^2 - b^2)$$

$$= kb^2 - kc^2 + kc^2 - ka^2 + ka^2 - kb^2$$

$$= 0 = \text{R.H.S}$$

(b)

Let the number be subtracted be x so, $(11-x) : (20-x) :: (26-x) : (50-x)$

$$\frac{11-x}{20-x} = \frac{26-x}{50-x}$$

$$(11-x)(50-x) = (26-x)(20-x)$$

$$550 - 11x - 50x + x^2 = 520 - 26x - 20x + x^2$$

$$550 - 61x = 520 - 46x$$

$$550 - 520 = -46x + 61x$$

$$30 = 15x$$

$$x = \frac{30}{15} = 2$$

Question: Solve the equation $3x - \frac{1}{x} = 2$ using formula method.

Or

Find the nature of the roots of the following equation,

$$6x^2 - x - 2 = 0$$

Solution:

(a)

$$3x - \frac{1}{x} = 2$$

$$3x^2 - 1 = 2x$$

$$3x^2 - 2x - 1 = 0$$

$$a = 3 \quad b = -2 \quad c = -1$$

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{2 \pm \sqrt{4 + 12}}{6} \\
 &= \frac{2 \pm 4}{6} \\
 &= 1, \frac{-2}{6}
 \end{aligned}$$

(b)

$$6x^2 - x - 2 = 0$$

$$a = 6 \quad b = -1 \quad c = -2$$

$$\Delta = b^2 - 4ac$$

$$= (-1)^2 - 4 \times 6 \times -2$$

$$= 1 + 48 = 49$$

Hence $\Delta > 0$

It means the root α and β are real and unequal.

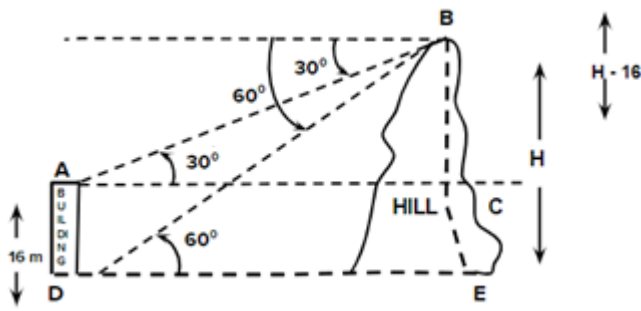
Question: From the top of the hill, the angle of depression of the top and the bottom of 16 m high building is 30° and 60° respectively. Find the height of the hill.

Or

An aeroplane is flying at a height of 8,000 m. The angle of depression of control tower of airport from the aeroplane is 30° . Find the horizontal distance between control tower and aeroplane

Solution:

(a)



let the height of hill be H

In $\triangle ABC$

$$\tan 30^\circ = \frac{BC}{AC}$$

$$\frac{1}{\sqrt{3}} = \frac{H-16}{AC}$$

$$AC = \sqrt{3}(H-16)$$

In $\triangle BDE$

$$\tan 60^\circ = \frac{BE}{DE}$$

$$DE = \frac{H}{\sqrt{3}}$$

From figure $AC = DE$

$$\sqrt{3}(H-16) = \frac{H}{\sqrt{3}}$$

$$3(H-16) = H$$

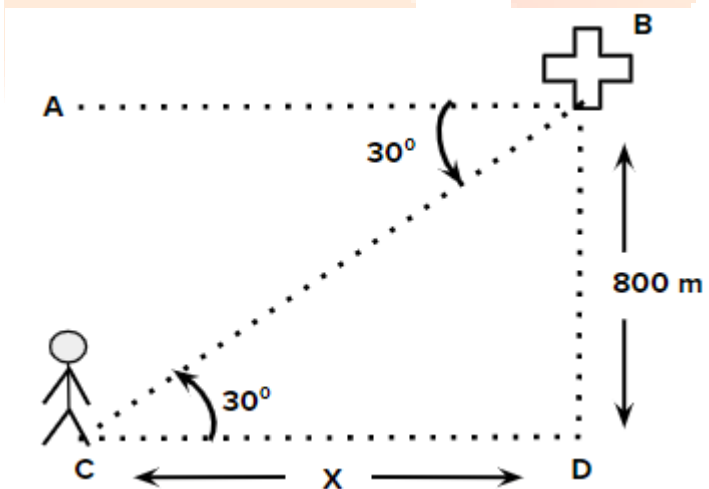
$$3H - 48 = H$$

$$2H = 48$$

$$H = 24$$

(b)

$\angle ABC = \angle BCD$ (interior angle)

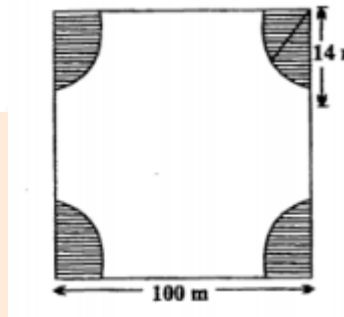


From figure

$$\tan 30 = \frac{8000}{x}$$

$$x = 8000\sqrt{3} \text{ m}$$

Question: A quadrant shaped flower bed is made of radius 14 m. in a square garden of side 100 m in all the four corners. Find the area of the remaining part of the square garden.

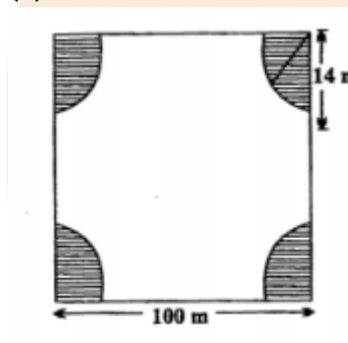


Or

A rocket in the form of closed cylinder from below, and upper part surmounted by cone of equal radius. The radius of cylinder is 2 m and height 21 m. The height of cone is 8.4 m. Find the volume of the rocket.

Solution:

(a)

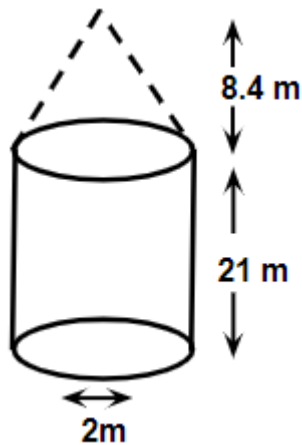


From figure

Remaining area = Ar. of flower bed – 4 × Ar. of quad flower

$$\begin{aligned}
 &= 100 \times 100 - 4 \times \frac{\pi}{4} \times r^2 \\
 &= 100 \times 100 - \frac{22}{7} \times 14 \times 14 \\
 &= 10000 - 616 \\
 &= 9384 \text{ m}^2
 \end{aligned}$$

(b)



Volume of Rocket = Volume of cone + Volume of cylinder

$$\begin{aligned} & \frac{1}{3} \pi r^2 h + \pi r^2 H \\ & \frac{1}{3} \times \frac{22}{7} \times 2 \times 2 \times 8.4 + \frac{22}{7} \times 2 \times 2 \times 21 \\ & = 35.2 + 264 \\ & = 299.2 \text{ m}^3 \end{aligned}$$

Question: The area of three adjacent faces of a cuboid are x, y and z. If the volume of cuboid is V, then prove that $V^2 = xyz$.

Or

A iron sphere of radius 8 cm melted then recasted into small sphere each of radius 1 cm. Find the number of small spheres.

Solution:

(a) let dimensions of cuboid be l, w, h

So, $x = l \times w$

$y = w \times h$

$z = h \times l$

R.H.S xyz

$lw \times wh \times hl$

$= (lwh)^2 = V^2 = \text{L.H.S}$

$\therefore V = lwh$

(b)

Total volume of bigger sphere = $\frac{4}{3} \pi r^3$

$$= \frac{4}{3} \pi \times 8^3$$

Volume of smaller sphere = $\frac{4}{3} \pi \times 1^3$

$$\text{No. of sphere} = \frac{\frac{4}{3}\pi \times 8^3}{\frac{4}{3}\pi \times 1^3} = 512$$

Question: Find cyclic factors: $ab(a - b) + bc(b - c) + ca(c - a)$.

Or

Multiply the rational expressions $\frac{x^2 - 7x + 10}{(x - 4)^2}$ and $\frac{x^2 - 7x + 12}{x - 5}$ express the product in its lowest terms.

Solution:

(a)

$$ab(a - b) + bc(b - c) + ca(c - a)$$

(Expand first and last term)

$$a^2b - ab^2 + bc(b - c) + c^2a - ca^2$$

(group terms containing a^2)

$$a^2(b - c) - a(b^2 - c^2) + bc(b - c)$$

$$(b - c)\{a^2 - a(b + c) + bc\}$$

(Taking common factor $b - c$ outside)

$$(b - c)\{a^2 - ab + ac + bc\}$$

$$(b - c)\{a(a - b) - c(a - b)\}$$

$$(b - c)(a - b)(a - c)$$

(b)

$$\frac{x^2 - 7x + 10}{(x - 4)^2} \times \frac{x^2 - 7x + 12}{x - 5}$$

$$\frac{x^2 - 5x - 2x + 10}{(x - 4)^2} \times \frac{x^2 - 4x - 3x + 12}{x - 5}$$

$$\frac{(x - 5)(x - 2)}{(x - 4)^2} \times \frac{(x - 4)(x - 3)}{(x - 5)}$$

$$= \frac{(x - 2)(x - 3)}{(x - 4)}$$

Question: If α, β are the roots of quadratic equation $3x^2 - 5x - 7 = 0$, then find the value of

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha}.$$

Or

The length of the side forming right angle of right triangle are x cm and $(x + 1)$ cm. If the area of the triangle is 10 cm^2 , then find the sides of the triangle.

Solution:

(a)

$$3x^2 - 5x - 7 = 0$$

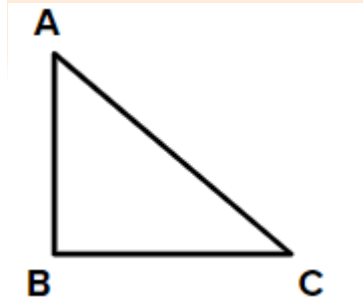
$$a = 3 \quad b = -5 \quad c = -7$$

$$\alpha + \beta = \frac{-b}{a} = \frac{5}{3}$$

$$\alpha\beta = \frac{c}{a} = \frac{-7}{3}$$

$$\begin{aligned} \frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{\left(\frac{5}{3}\right)^2 + 2 \times \frac{7}{3}}{\frac{-7}{3}} = \frac{\frac{25}{9} + \frac{14}{3}}{\frac{-7}{3}} \\ &= \frac{25 + 42}{9} \times \frac{3}{-7} \\ &= \frac{67}{-21} \end{aligned}$$

(b)



Let AB be $x + 1$

BC be x of right angled

$$\text{Area of } \triangle ABC = \frac{1}{2} \times AB \times BC$$

$$10 = \frac{1}{2} \times (x+1)x$$

$$20 = x^2 + x$$

$$0 = x^2 + x - 20$$

$$= x^2 + 5x - 4x - 20$$

$$= x(x+5) - 4(x+5)$$

$$= (x+5)(x-4)$$

$$x = -5 \text{ (or) } x = 4$$

Hence, sides are 4 cm and 5 cm.

Question: Find the compound interest on Rs.2,000 at the rate of interest 4% per annum for 2 years.

Or

A sewing machine is available for Rs. 1,600 cash or for Rs. 1,200 cash down payment and Rs. 460 to be paid after 6 months. Find the rate of interest charged under the instalment plan.

Solution:

(a) Principal = 2000 Rate = 4% Time = 2

$$\begin{aligned}
 A &= P \left(1 + \frac{r}{100} \right)^n \\
 &= 2000 \left(1 + \frac{4}{100} \right)^2 \\
 &= 2000 \times \frac{104}{100} \times \frac{104}{100} = 2163.2 \text{ Rs} \\
 \text{CI} &= A - P \\
 &= 2163.2 - 2000 \\
 &= 163.2 \text{ Rs}
 \end{aligned}$$

(b)

Cash price of sewing machine = 1600 Rs

Down payment = 1200 Rs

Balance due = 1600 – 1200 = 400 Rs

No. of instalment = 1

Amount of instalment = 460 Rs

Interest paid = 460 – 400 = 60 Rs

$$\begin{aligned}
 \text{SI} &= \frac{PRT}{100} \\
 60 &= \frac{1600 \times R \times 6}{100 \times 12} \\
 R &= 7.5 \%
 \end{aligned}$$

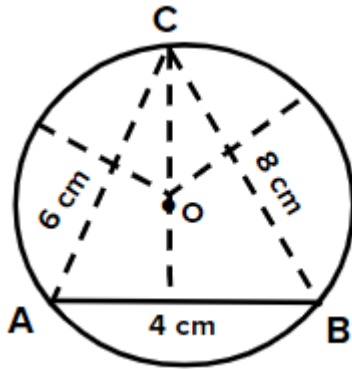
Question: The side of the triangle are 4 cm, 6 cm and 8 cm. Draw the circumcircle of the triangle and write the steps of construction.

Or

Construct a cyclic quadrilateral in which AC = 6 cm, $\angle B = 90^\circ$, AB = 3 cm and AD = 4 cm. Write the steps of construction also.

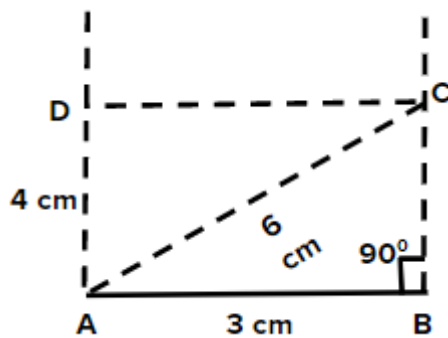
Solution:

(a)



- (i) Draw $AB = 4$ cm
- (ii) Take arc of 6 cm and 8 cm draw arcs
- (iii) Join A,B,C to get $\triangle ABC$
- (iv) Draw perpendicular bisector of each side AB, BC, CA
- (v) The point where perpendicular bisector meet name O
- (vi) Draw circumcircle.

(b)



- (i) Draw $AB = 3$ cm
- (ii) Draw $180^\circ = \angle ABC + \angle ADC$
 $\angle B + \angle D = 180^\circ$
- (iii) From A, draw Arc of 6 cm cutting BC
- (iv) Draw $\angle A = 90^\circ$
- (v) cut Arc of 4 cm
- (vi) Join to get ABCD quad.

Question: Prove that: $\sin^2 \theta + \cos^2 \theta = 1$

Or

Show that whether the following is identity or not:

$$\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{\sec \theta + 1}{\sec \theta - 1}$$

Solution:

(a)

Consider triangle of height L, Base B, Hypotenuses H

$$\sin \theta = \frac{L}{H} \quad \cos \theta = \frac{B}{H}$$

Also, $L^2 + B^2 = H^2$

L.H.S $\sin^2 \theta + \cos^2 \theta$

$$\frac{L^2}{H^2} + \frac{B^2}{H^2} = \frac{L^2 + B^2}{H^2} = \frac{H^2}{H^2} = 1 \quad \text{R.H.S}$$

(b)

$$\begin{aligned} \text{L.H.S} &= \frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} \\ &= \frac{\frac{\sin \theta}{\cos \theta} + \sin \theta}{\frac{\sin \theta}{\cos \theta} - \sin \theta} \\ &= \frac{\sin \theta \left(\frac{1}{\cos \theta} + 1 \right)}{\sin \theta \left(\frac{1}{\cos \theta} - 1 \right)} \\ &= \frac{\sec \theta + 1}{\sec \theta - 1} = \text{R.H.S} \end{aligned}$$

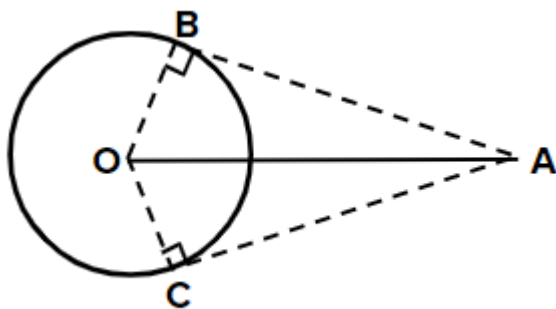
Question: Prove that the length of two tangents drawn from an external point to a circle are equal.

Or

Write the definition of cyclic quadrilateral. Prove that the sum of pairs of opposite angles of a cyclic quadrilateral is 180°

Solution:

(a)



In $\triangle ABO$ and $\triangle ACO$

$AO = AO$ (common)

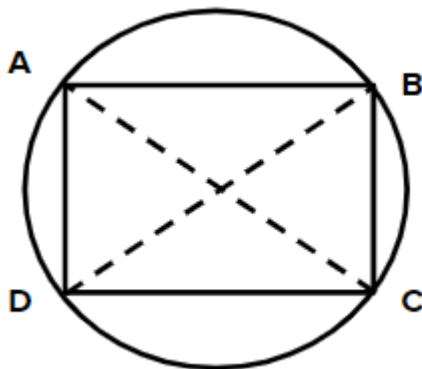
$OB = OC$ (Radius)

$\angle OBA = \angle OCA$ (90°)

By SAS $\triangle ABO \cong \triangle ACO$

Hence, $AB = AC$ by c. p. c. t

(b)



A cyclic quadrilateral is a quadrilateral whose vertices all lie on a single circle. Thus circle is called circumcircle and vertices are called concyclic.

To Prove: $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$

Join AC and BD

$$\angle ACB = \angle ADB \text{ (angle in same segment)}$$

$$\angle BAC = \angle BDC \text{ (angle in same segment)}$$

$$\angle ACB + \angle BAC = \angle ADB + \angle BDC$$

$$\angle ABC + \angle ACB + \angle BAC = \angle ADC + \angle ABC \text{ (add } \angle ABC \text{ on both)}$$

$$\angle B + \angle D = 180^\circ$$

Similarly $\angle A + \angle C = 180^\circ$

Question: Compute the mean by short cut method of the following frequency distribution:

Marks Obtained	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Number of students	6	8	13	7	3	2	1

Solution:

Item	Quantity (units)	Price (in Rs.) per unit	
		Year 1996	Year 1999
A	8		
B	12	22	25
C	5	35	40
D	15	25	30
E	10	20	25
		15	20

Marks	Mid Value	f	di = xi – A	f di
10 – 20	15	6	– 30	– 180
20 – 30	25	8	– 20	– 160
30 – 40	35	13	– 10	– 130
40 – 50	45 – A	7	0	0
50 – 60	55	3	10	30
60 – 70	65	2	20	40
70 – 80	75	1	30	30
	$\sum f = 40$			$\sum f di = - 370$

$$\text{As } \bar{x} = A + \frac{\sum f di}{\sum f} \quad (\text{where } A = \text{assumed mean and } \bar{x} \text{ mean})$$

$$= 45 + \frac{-370}{40}$$

$$= 35.75$$